

```
import pandas as pd
```

```
import os
os.getcwd()
```

```
'C:\\Users\\lenovo\\OneDrive\\Desktop'
```

```
os.listdir()
```

```
['.ipynb_checkpoints',
 '3raisyll.pdf',
 'desktop.ini',
 'E-Commerce',
 'E-commerce.97fd8198-eccd-4848-8839-2b1999225b3e',
 'EXCEL SHEET(e-COMMERCE).xlsx',
 'NumPy',
 'numpy.pdf',
 'PANDA's",
 'Pandas.pdf',
 'Untitled.ipynb']
```

```
os.listdir("E-Commerce")
```

```
['EXCEL SHEET(e-COMMERCE) updated answers.xlsx',
 'EXCEL SHEET(e-COMMERCE).xlsx']
```

```
%pip install openpyxl
```

Requirement already satisfied: openpyxl in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.1.5)
 Requirement already satisfied: et-xmlfile in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packages (from openpyxl)
 Note: you may need to restart the kernel to use updated packages.

```
[notice] A new release of pip is available: 26.1.1 -> 26.2.1
```

```
[notice] To update, run: C:\Users\lenovo\AppData\Local\Programs\Python\Python314\python.exe -m pip install --upgrade pip
```

```
df=pd.read_excel("E-Commerce/EXCEL SHEET(e-COMMERCE).xlsx")
```

```
df.head()
```

	order_id	order_date	ship_date	customer	manufactory	product_name	segment	category	subcategory	region	zip	city
0	US-2020-103800	2019-01-03	2019-01-07	Darren Powers	Message Book	Message Book, Wirebound, Four 5 1/2" X 4" Form...	Consumer	Office Supplies	Paper	Central	77095	Houston
1	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	GBC	GBC Standard Plastic Binding Systems Combs	Home Office	Office Supplies	Binders	Central	60540	Naperville
2	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	Avery	Avery 508	Home Office	Office Supplies	Labels	Central	60540	Naperville
3	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	SAFCO	SAFCO Boltless Steel Shelving	Home Office	Office Supplies	Storage	Central	60540	Naperville
4	US-2020-141817	2019-01-05	2019-01-12	Mick Brown	Avery	Avery Hi-Liter EverBold Pen Style Fluorescent ...	Consumer	Office Supplies	Art	East	19143	Philadelphia

```
df.tail()
```

	order_id	order_date	ship_date	customer	manufactory	product_name	segment	category	subcategory	region	zip	
9989	US-2023-126221	2022-12-30	2023-01-05	Chuck Clark	Eureka	Eureka The Boss Plus 12-Amp Hard Box Upright V...	Home Office	Office Supplies	Appliances	Central	47201	Colu
9990	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Other	Bush Westfield Collection Bookcases, Fully Ass...	Consumer	Furniture	Bookcases	East	10009	Ne
9991	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Wilson Jones	Wilson Jones Legal Size Ring Binders	Consumer	Office Supplies	Binders	East	10009	Ne
9992	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Other	Gear Head AU3700S Headset	Consumer	Technology	Phones	East	10009	Ne
9993	US-2023-156720	2022-12-30	2023-01-03	Jill Matthias	Other	Bagged Rubber Bands	Consumer	Office Supplies	Fasteners	West	80538	Lo

df.describe()

	order_date	ship_date	zip	discount	profit	quantity	sales	profit_margin
count	9994	9994	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	2021-04-30 05:17:08.056834	2021-05-04 04:17:20.304182	55190.371723	0.156203	28.656896	3.789574	229.858001	0.120314
min	2019-01-03 00:00:00	2019-01-07 00:00:00	1040.000000	0.000000	-6599.978000	1.000000	0.444000	-2.750000
25%	2020-05-23 00:00:00	2020-05-27 00:00:00	23223.000000	0.000000	1.728750	2.000000	17.280000	0.075000
50%	2021-06-26 00:00:00	2021-06-29 00:00:00	56430.500000	0.200000	8.666500	3.000000	54.490000	0.270000
75%	2022-05-14 00:00:00	2022-05-18 00:00:00	90008.000000	0.200000	29.364000	5.000000	209.940000	0.362500
max	2022-12-30 00:00:00	2023-01-05 00:00:00	99301.000000	0.800000	8399.976000	14.000000	22638.480000	0.500000
...	...	...	...	...	...	...	...	...

df.shape

(9994, 19)

df.info

<bound method DataFrame.info of				order_id	order_date	ship_date	customer	manufactory	\	
0	US-2020-103800	2019-01-03	2019-01-07	Darren Powers	Message Book					
1	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	GBC					
2	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	Avery					
3	US-2020-112326	2019-01-04	2019-01-08	Phillina Ober	SAFCO					
4	US-2020-141817	2019-01-05	2019-01-12	Mick Brown	Avery					
...	...	...	...	...	...					
9989	US-2023-126221	2022-12-30	2023-01-05	Chuck Clark	Eureka					
9990	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Other					
9991	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Wilson Jones					
9992	US-2023-143259	2022-12-30	2023-01-03	Patrick O'Donnell	Other					
9993	US-2023-156720	2022-12-30	2023-01-03	Jill Matthias	Other					
				product_name	segment	\				
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...			Consumer						
1	GBC Standard Plastic Binding Systems Combs			Home Office						
2	Avery 508			Home Office						
3	SAFCO Boltless Steel Shelving			Home Office						
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...			Consumer						
...	...			...						
9989	Eureka The Boss Plus 12-Amp Hard Box Upright V...			Home Office						
9990	Bush Westfield Collection Bookcases, Fully Ass...			Consumer						
9991	Wilson Jones Legal Size Ring Binders			Consumer						
9992	Gear Head AU3700S Headset			Consumer						
9993	Bagged Rubber Bands			Consumer						
				category	subcategory	region	zip	city	\	
0	Office Supplies		Paper	Central	77095	Houston				
1	Office Supplies		Binders	Central	60540	Naperville				
2	Office Supplies		Labels	Central	60540	Naperville				
3	Office Supplies		Storage	Central	60540	Naperville				

```

4      Office Supplies      Art      East 19143 Philadelphia
...      ...      ...      ...      ...
9989 Office Supplies Appliances Central 47201 Columbus
9990      Furniture Bookcases East 10009 New York City
9991 Office Supplies Binders East 10009 New York City
9992      Technology Phones East 10009 New York City
9993 Office Supplies Fasteners West 80538 Loveland

      state      country discount profit quantity sales \
0      Texas United States 0.2 5.5512 2 16.448
1      Illinois United States 0.8 -5.4870 2 3.540
2      Illinois United States 0.2 4.2717 3 11.784
3      Illinois United States 0.2 -64.7748 3 272.736
4      Pennsylvania United States 0.2 4.8840 3 19.536
...      ...      ...      ...      ...
9989      Indiana United States 0.0 56.5110 2 209.300
9990      New York United States 0.2 12.1176 4 323.136
9991      New York United States 0.2 19.7910 3 52.776
9992      New York United States 0.0 2.7279 7 90.930
9993      Colorado United States 0.2 -0.6048 3 3.024

      profit_margin
0      0.3375
1      -1.5500
2      0.3625
3      -0.2375
4      0.2500

```

```
df.dtypes
```

```

order_id      str
order_date    datetime64[us]
ship_date     datetime64[us]
customer      str
manufactory   str
product_name  str
segment      str
category      str
subcategory   str
region        str
zip           int64
city          str
state         str
country       str
discount      float64
profit        float64
quantity      int64
sales         float64
profit_margin float64
dtype: object

```

```
df.T
```

	0	1	2	3	4	5	6	7	8	9	...
order_id	US-2020-103800	US-2020-112326	US-2020-112326	US-2020-112326	US-2020-141817	US-2020-106054	US-2020-130813	US-2020-167199	US-2020-167199	US-2020-167199	...
order_date	2019-01-03 00:00:00	2019-01-04 00:00:00	2019-01-04 00:00:00	2019-01-04 00:00:00	2019-01-05 00:00:00	2019-01-06 00:00:00	2019-01-06 00:00:00	2019-01-06 00:00:00	2019-01-06 00:00:00	2019-01-06 00:00:00	...
ship_date	2019-01-07 00:00:00	2019-01-08 00:00:00	2019-01-08 00:00:00	2019-01-08 00:00:00	2019-01-12 00:00:00	2019-01-07 00:00:00	2019-01-08 00:00:00	2019-01-10 00:00:00	2019-01-10 00:00:00	2019-01-10 00:00:00	...
customer	Darren Powers	Phillina Ober	Phillina Ober	Phillina Ober	Mick Brown	Jack O'Briant	Lycoris Saunders	Maria Etezadi	Maria Etezadi	Maria Etezadi	...
manufactory	Message Book	GBC	Avery	SAFCO	Avery	Dixon	Xerox	Global	Rogers	Ibico	...
product_name	Message Book, Wirebound, Four 5 1/2" X 4" Form...	GBC Standard Plastic Binding Systems Combs	Avery 508	SAFCO Boltless Steel Shelving	Avery Hi-Liter EverBold Pen Style Fluorescent ...	Dixon Prang Watercolor Pencils, 10-Color Set w...	Xerox 225	Global Deluxe High-Back Manager's Chair	Rogers Handheld Barrel Pencil Sharpener	Ibico Hi-Tech Manual Binding System	...
segment	Consumer	Home Office	Home Office	Home Office	Consumer	Corporate	Consumer	Home Office	Home Office	Home Office	...
category	Office Supplies	Office Supplies	Office Supplies	Office Supplies	Office Supplies	Office Supplies	Office Supplies	Furniture	Office Supplies	Office Supplies	...
subcategory	Paper	Binders	Labels	Storage	Art	Art	Paper	Chairs	Art	Binders	...
region	Central	Central	Central	Central	East	South	West	South	South	South	...
zip	77095	60540	60540	60540	19143	30605	90049	42420	42420	42420	...
city	Houston	Naperville	Naperville	Naperville	Philadelphia	Athens	Los Angeles	Henderson	Henderson	Henderson	...
state	Texas	Illinois	Illinois	Illinois	Pennsylvania	Georgia	California	Kentucky	Kentucky	Kentucky	...
country	United States	United States	United States	United States	United States	United States	United States	United States	United States	United States	...
discount	0.2	0.8	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	...
profit	5.5512	-5.487	4.2717	-64.7748	4.884	5.2398	9.3312	746.4078	1.4796	274.491	...
quantity	2	2	3	3	3	3	3	9	2	2	...
sales	16.448	3.54	11.784	272.736	19.536	12.78	19.44	2573.82	5.48	609.98	...
profit_margin	0.3375	-1.55	0.3625	-0.2375	0.25	0.41	0.48	0.29	0.27	0.45	...

19 rows × 9994 columns

df.loc

<pandas.core.indexing._LocIndexer at 0x2b3fd4f3330>

df.iloc

<pandas.core.indexing._iLocIndexer at 0x2b3fd2ed980>

df.isnull().sum()

order_id	0
order_date	0
ship_date	0
customer	0
manufactory	0
product_name	0
segment	0
category	0
subcategory	0
region	0
zip	0
city	0
state	0
country	0

```
discount      0
profit        0
quantity      0
sales         0
profit_margin 0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(1)
```

```
df[df.duplicated()]
```

	order_id	order_date	ship_date	customer	manufactory	product_name	segment	category	subcategory	region	zip	city
386	US-2020-150119	2019-04-23	2019-04-27	Laurel Beltran	Global	Global Leather Highback Executive Chair with P...	Home Office	Furniture	Chairs	East	43229	Columbu

```
df[df.duplicated(keep=False)]
```

	order_id	order_date	ship_date	customer	manufactory	product_name	segment	category	subcategory	region	zip	city
385	US-2020-150119	2019-04-23	2019-04-27	Laurel Beltran	Global	Global Leather Highback Executive Chair with P...	Home Office	Furniture	Chairs	East	43229	Columbu
386	US-2020-150119	2019-04-23	2019-04-27	Laurel Beltran	Global	Global Leather Highback Executive Chair with P...	Home Office	Furniture	Chairs	East	43229	Columbu

```
df=df.drop_duplicates()
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df["sales"].sum()
```

```
np.float64(2296919.4883)
```

```
df.groupby("category")["sales"].sum()
```

```
category
Furniture      741718.4233
Office Supplies 719047.0320
Technology     836154.0330
Name: sales, dtype: float64
```

```
df.groupby("region")["sales"].sum()
```

```
region
Central      501239.8908
East         678499.8680
South        391721.9050
West         725457.8245
Name: sales, dtype: float64
```

```
df["year"]=df["order_date"].dt.year
```

```
df.groupby("year")["sales"].sum()
```

```
year
2019      483966.1261
2020      470532.5090
2021      609205.5980
2022      733215.2552
Name: sales, dtype: float64
```

```
df.groupby("category")["profit"].sum().sort_values(ascending=False)
```

```
category
Technology      145454.9481
Office Supplies  122490.8008
Furniture        18463.3316
Name: profit, dtype: float64
```

```
df.groupby("subcategory")["loss"].sum().sort_values(ascending=False)
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[35], line 1
----> 1 df.groupby("subcategory")["loss"].sum().sort_values(ascending=False)

File ~\AppData\Local\Programs\Python\Python314\Lib\site-packages\pandas\core\groupby\generic.py:2938, in
DataFrameGroupBy.__getitem__(self, key)
    2931 if isinstance(key, tuple) and len(key) > 1:
    2932     # if len == 1, then it becomes a SeriesGroupBy and this is actually
    2933     # valid syntax, so don't raise
    2934     raise ValueError(
    2935         "Cannot subset columns with a tuple with more than one element. "
    2936         "Use a list instead."
    2937     )
-> 2938 return super().__getitem__(key)

File ~\AppData\Local\Programs\Python\Python314\Lib\site-packages\pandas\core\base.py:228, in SelectionMixin.__getitem__(self, key)
    226 else:
    227     if key not in self.obj:
-> 228         raise KeyError(f"Column not found: {key}")
    229     ndim = self.obj[key].ndim
    230     return self._getitem(key, ndim=ndim)

KeyError: 'Column not found: loss'
```

```
sales_by_category=df.groupby("category")["sales"].sum()
```

```
sales_by_category.plot(kind="bar",title="Sales by Category")
```

```
-----
ImportError                                Traceback (most recent call last)
Cell In[37], line 1
----> 1 sales_by_category.plot(kind="bar",title="Sales by Category")

File ~\AppData\Local\Programs\Python\Python314\Lib\site-packages\pandas\plotting\_core.py:1105, in PlotAccessor.__call__(self, *args, **kwargs)
    1104 def __call__(self, *args, **kwargs):
-> 1105     plot_backend = _get_plot_backend(kwargs.pop("backend", None))
    1107     x, y, kind, kwargs = self._get_call_args(
    1108         plot_backend.__name__, self._parent, args, kwargs
    1109     )
    1111     kind = self._kind_aliases.get(kind, kind)

File ~\AppData\Local\Programs\Python\Python314\Lib\site-packages\pandas\plotting\_core.py:2251, in _get_plot_backend(backend)
    2248 if backend_str in _backends:
    2249     return _backends[backend_str]
-> 2251 module = _load_backend(backend_str)
    2252 _backends[backend_str] = module
    2253 return module

File ~\AppData\Local\Programs\Python\Python314\Lib\site-packages\pandas\plotting\_core.py:2183, in _load_backend(backend)
    2181     module = importlib.import_module("pandas.plotting._matplotlib")
    2182     except ImportError:
-> 2183         raise ImportError(
    2184             "matplotlib is required for plotting when the "
    2185             "default backend 'matplotlib' is selected."
    2186         ) from None
    2187     return module
    2189 found_backend = False

ImportError: matplotlib is required for plotting when the default backend "matplotlib" is selected.
```

```
%pip install matplotlib
```

```
Collecting matplotlib
  Using cached matplotlib-3.11.2-cp314-cp314-win_amd64.whl.metadata (80 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Using cached contourpy-1.4.0-cp314-cp314-win_amd64.whl.metadata (3.8 kB)
```

```

Collecting cycler>=0.10 (from matplotlib)
  Using cached cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl.metadata (132 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Using cached kiwisolver-1.5.1-cp314-cp314-win_amd64.whl.metadata (5.2 kB)
Requirement already satisfied: numpy>=1.25 in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packages (from r
Requirement already satisfied: packaging>=20.0 in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packages (f
Collecting pillow>=9 (from matplotlib)
  Using cached pillow-12.3.0-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packag
Requirement already satisfied: six>=1.5 in C:\Users\lenovo\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pytl
Using cached matplotlib-3.11.2-cp314-cp314-win_amd64.whl (9.5 MB)
Using cached contourpy-1.4.0-cp314-cp314-win_amd64.whl (240 kB)
Using cached cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl (2.5 MB)
----- 0.0/2.5 MB ? eta -:-:--
----- 0.3/2.5 MB ? eta -:-:--
----- 0.5/2.5 MB 1.3 MB/s eta 0:00:02
----- 1.0/2.5 MB 1.7 MB/s eta 0:00:01
----- 1.3/2.5 MB 1.8 MB/s eta 0:00:01
----- 1.8/2.5 MB 1.8 MB/s eta 0:00:01
----- 2.4/2.5 MB 1.9 MB/s eta 0:00:01
----- 2.5/2.5 MB 1.8 MB/s 0:00:01
Using cached kiwisolver-1.5.1-cp314-cp314-win_amd64.whl (72 kB)
Using cached pillow-12.3.0-cp314-cp314-win_amd64.whl (7.2 MB)
Downloading pyparsing-3.3.3-py3-none-any.whl (126 kB)
Installing collected packages: pyparsing, pillow, kiwisolver, fonttools, cycler, contourpy, matplotlib

```

```

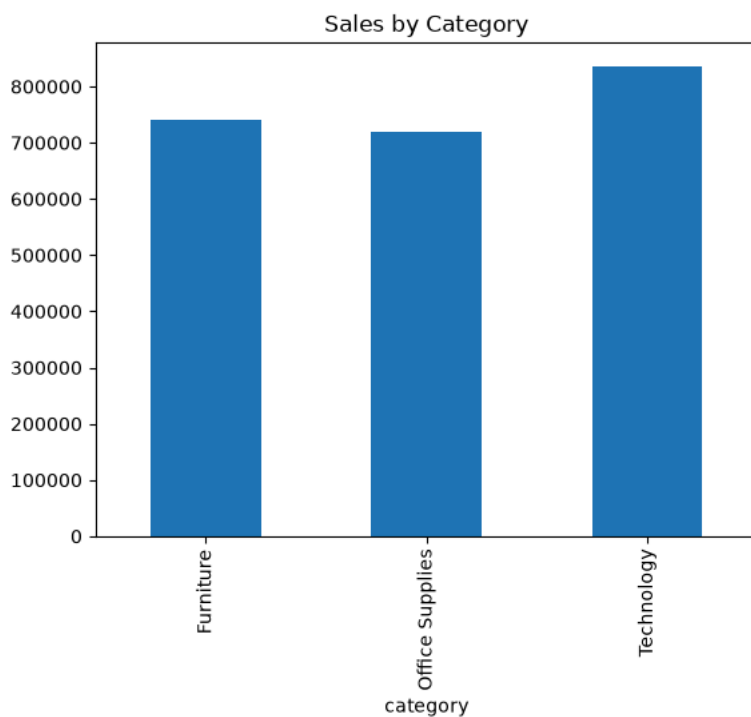
----- 0/7 [pyparsing]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 1/7 [pillow]
----- 2/7 [kiwisolver]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]
----- 3/7 [fonttools]

```

```
sales_by_category=df.groupby("category")["sales"].sum()
```

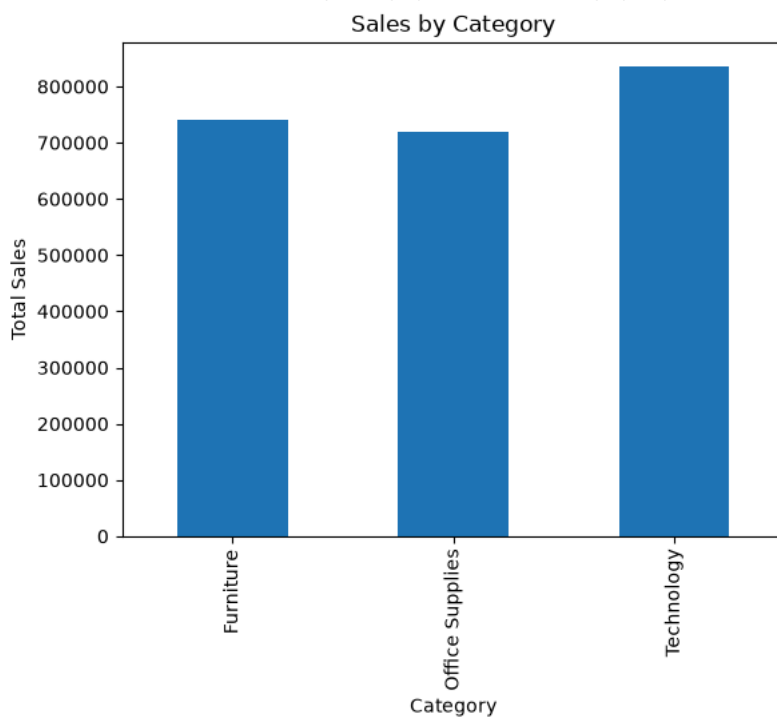
```
sales_by_category.plot(kind="bar",title="Sales by Category")
```

Matplotlib is building the font cache; this may take a moment.
<Axes: title={'center': 'Sales by Category'}, xlabel='category'>



```
sales_by_category.plot(kind="bar",title="Sales by Category",xlabel="Category",ylabel="Total Sales")
```

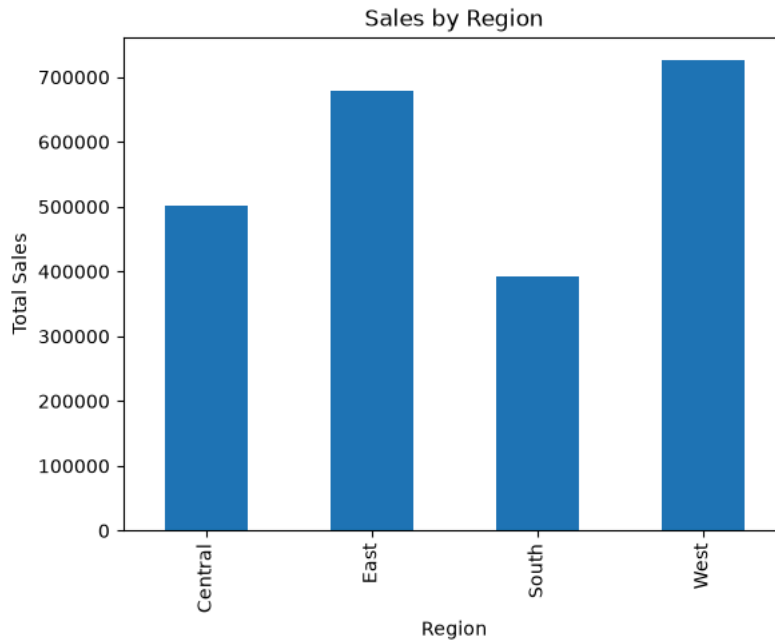
<Axes: title={'center': 'Sales by Category'}, xlabel='Category', ylabel='Total Sales'>



```
sales_by_region=df.groupby("region")["sales"].sum()
```

```
sales_by_region.plot(kind="bar",title="Sales by Region",xlabel="Region",ylabel="Total Sales")
```

```
<Axes: title={'center': 'Sales by Region'}, xlabel='Region', ylabel='Total Sales'>
```



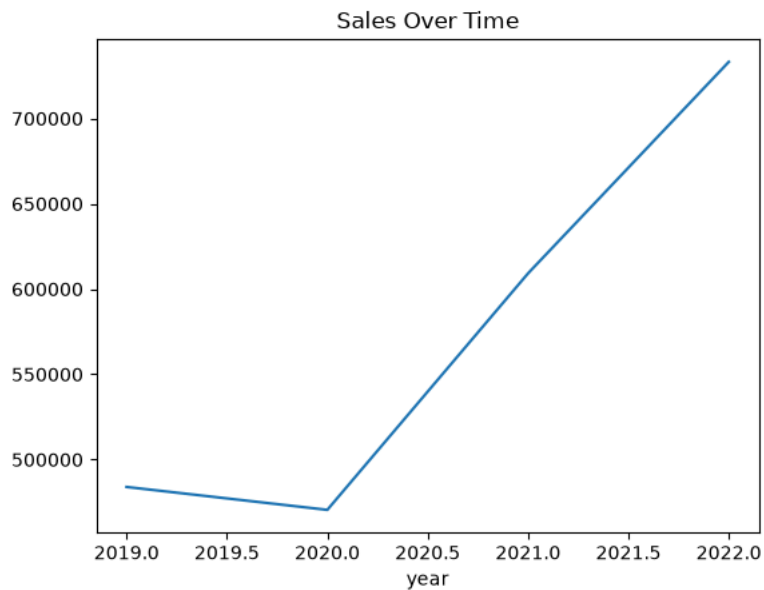
```
df["year"]=df["order_date"].dt.year
```

```
a=df.groupby("year")["sales"].sum()
```

```
sales_by_year=df.groupby("year")["sales"].sum()
```

```
sales_by_year.plot(kind="line",title="Sales Over Time")
```

```
<Axes: title={'center': 'Sales Over Time'}, xlabel='year'>
```



```
profit_by_category=df.groupby("category")["profit"].sum()
```

```
profit_by_category.plot(kind="bar",title="Profit by Category",xlabel="Category",ylabel="Total Profit")
```

<Axes: title={'center': 'Profit by Category'}, xlabel='Category', ylabel='Total Profit'>



```
profit_by_subcategory=df.groupby("subcategory")["profit"].sum()
```

```
profit_by_subcategory.plot(kind="bar",title="Profit by Subcategory",xlabel="Subcategory",ylabel="Total Profit")
```

<Axes: title={'center': 'Profit by Subcategory'}, xlabel='Subcategory', ylabel='Total Profit'>

